Experiment 1 on 28-10-2026

learned post-training calibration scaling of spatial regularisation maps

Stage I: exploratory four-image scale sweep

Reproducible experimental report

28 October 2026, 13:11 UTC

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1 Executive summary

This report analyses a four-image CIGS PCM experiment with 12 held-out noisy reconstruction data (three noise realisations per image). The four ground-truth images, not the 12 noise realisations, are the relevant units of structural diversity.

Relative to using the original learned parameter map at $c = 1$, the test-set posthoc per-datum oracle gain is 4.004 dB in mean PSNR; a single calibration-selected multiplier gains 4.004 dB. The remaining test oracle versus fixed multiplier opportunity is 0.000 dB. These are descriptive findings on four images only; the oracle is not a deployable predictor.

Stage I deliberately does NOT train a scaling head. Its role is to measure whether the potential improvement, label stability, near-optimal scale sets and global-quality/line-preservation trade-off justify Stage II: Approach A with 50 separate training images across five categories, 25 validation images, and a final test on the same four CIGS structures but new noise seeds.

2 Experiment design and reproducibility

Given noisy data $\boldsymbol{y} = A\boldsymbol{x}^\star + \boldsymbol{\varepsilon}$ and one predicted learned TV map $\boldsymbol{\Lambda}_\theta(\boldsymbol{y})$, the only varied parameter is a positive scalar c:

$$\boldsymbol{x}_c = \mathrm{PDHG}_K(\boldsymbol{y}, c\boldsymbol{\Lambda}_\theta(\boldsymbol{y})).$$

The map is predicted *once* for each datum and frozen across the entire c sweep. There are no outer Bregman iterations. Finite-step PDHG is not an exact optimisation solution by default.

Property	Recorded configuration
Experiment	Experiment 1; fixed-map scaling only
Images	CIGS 0674, CIGS 0985, CIGS 0917, CIGS 0864
Sampling	20 percent normalised Hadamard; coarse J=6
Noise	LION NPL simulation; 2 calibration seeds + 3 held-out seeds
Coarse multipliers	0.125, 0.25, 0.5, 1, 2, 4, 8, 16, 32, 64
Refinement	Logarithmic factors around calibration optima; from calibration seeds only
Model	old
Actual PDHG depth	256
Frozen checkpoint	best_val_loss_s0.20_cJ6.pt
Checkpoint SHA256	4da3f177ece6dc6f2397...
LION Git revision	ce5dcb4780780d9779...
No. of coarse reconstructions	200
No. of refined reconstructions	140

Training/evaluation separation: Stage I is PRETRAINING exploratory analysis. Calibration seed ground truths can select scales; the three designated test noise seeds never select the pooled, per-image, or leave-one-image-out scale. Posthoc test oracle values are plotted ONLY as upper bounds. The per-image calibrated selector is an image-specific oracle requiring the same image in calibration; it is not a blind per-datum scaling system.

3 How Approach A target quality is assessed

For noisy datum i, s and discrete tested scale set $\mathcal{S}$, define $Q_{is}(c) = \mathrm{PSNR}(\boldsymbol{x}_{is,c}, \boldsymbol{x}_i^\star)$, $Q_{is}^\star = \max_{c\in\mathcal{S}} Q_{is}(c)$, and

$$c_{is}^\star \in \arg\max_{c\in\mathcal{S}} Q_{is}(c).$$

A hard $c^\star$ is unstable when several scales give nearly identical PSNR. We also retain the complete sampled near-optimal *set*

$$\mathcal{C}_{is,\epsilon} = \{c \in \mathcal{S} : Q_{is}(c) \geq Q^\star_{is} - \epsilon\}, \qquad \epsilon = 0.1 \text{ dB},$$

which may be disconnected. The plot of its minimum-to-maximum span is a visual hull, not a claim that every intervening continuous scale is acceptable.

For future offline Approach A training, the soft label over measured scales is

$$p_{is}(c_j) = \frac{\exp([Q_{is}(c_j) - Q^\star_{is}]/T)}{\sum_\ell \exp([Q_{is}(c_\ell) - Q^\star_{is}]/T)}, \qquad T = 0.15 \text{ dB}.$$

Prediction by squared log-scale error, prediction of an acceptable interval, and soft-label classification should be compared later using *reconstruction regret*, not just oracle-label regression MSE.

4 Global-calibration results: tables

Pooled PSNR-selected multiplier from calibration ground truths: $c_{\text{pooled}} = 8.0000$. The table below shows scale values, not image-quality scores.

CIGS	PSNR c	SSIM c	HaarPSI c	line-aware c	LOIO c
0674	8.0000	8.0000	5.6569	4.0000	8.0000
0985	9.5137	8.0000	5.6569	4.7568	8.0000
0917	8.0000	8.0000	5.6569	4.0000	8.0000
0864	8.0000	6.7272	4.7568	4.0000	8.0000

Test-set PSNR improvements in dB relative to $c = 1$, averaged first across the three test seeds for each image. LOIO means the pooled scale was calibrated without this image.

CIGS	Pooled gain	LOIO gain	Oracle gain	Oracle minus pooled
0674	4.240	4.240	4.240	0.000
0985	3.984	3.984	3.985	0.001
0917	3.862	3.862	3.862	0.000
0864	3.929	3.929	3.929	0.000
Four-image mean	4.004	4.004	4.004	0.000

A large oracle-minus-pooled gap may justify an adaptive scalar predictor, but only if the oracle scale is stable and predictable from the noisy input. A large oracle gain alone does not establish that predictability.

5 Quality and line-preservation figures

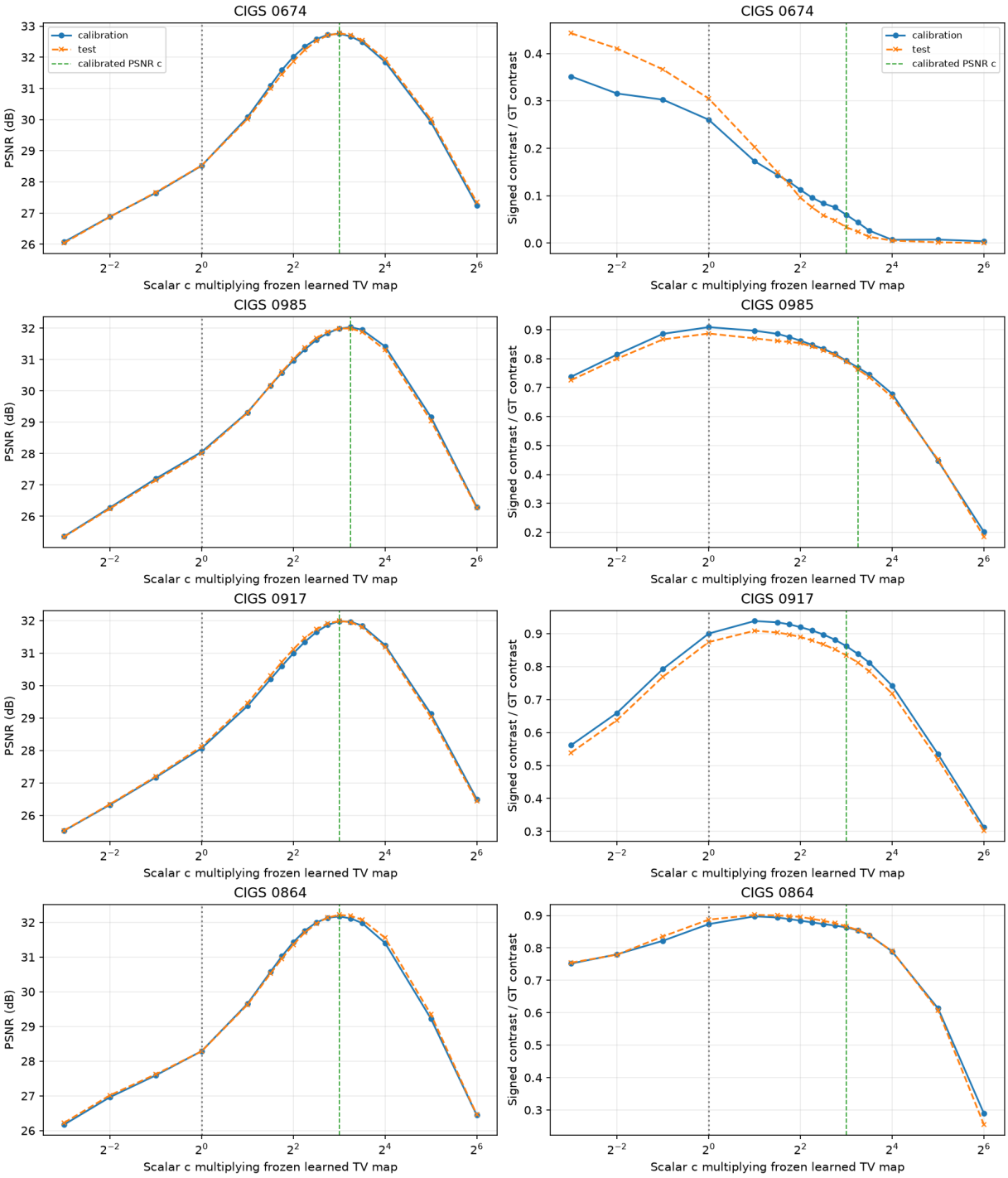


Figure 1: PSNR and signed contrast retention versus scale. Dotted c=1 and calibration-optimum markers reveal whether a better global metric coincides with diminished annotated-line contrast. Test curves are displayed, not used for calibration.

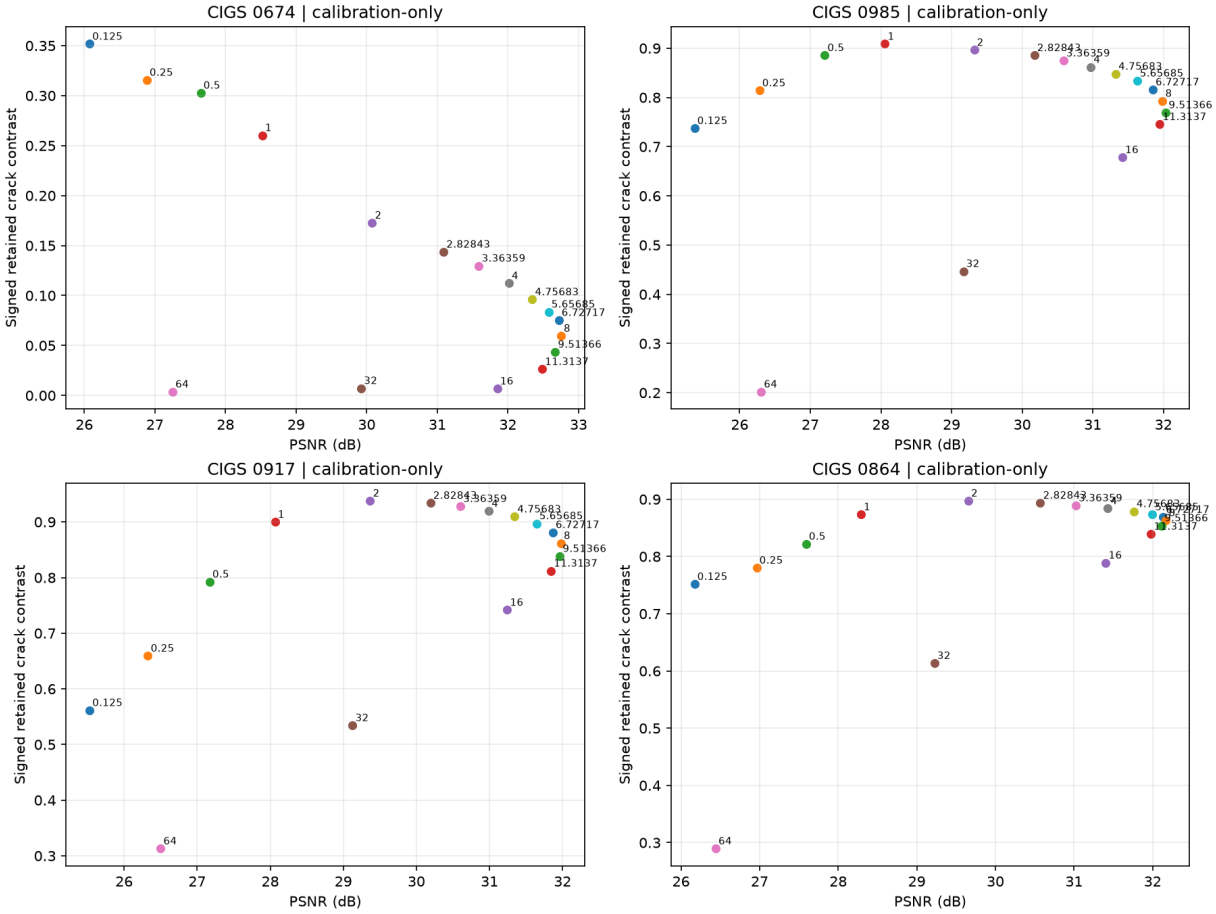


Figure 2: PSNR versus annotated-line contrast for each CIGS image. The labelled points are scales from calibration noise; visualise the global-quality versus line-preservation trade-off rather than collapsing them into one score.

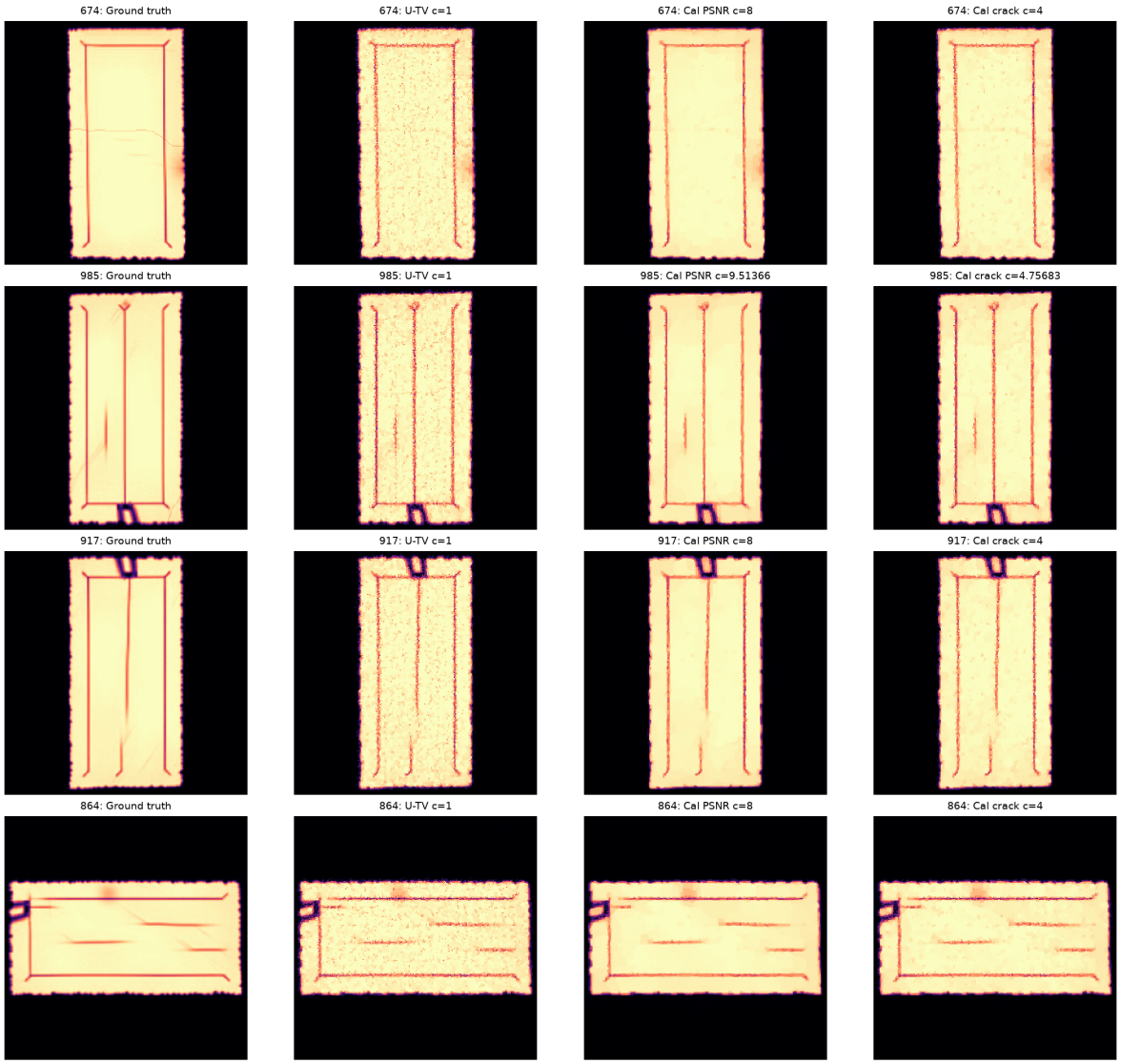


Figure 3: Full-image reconstruction overview. All panels use a common PCM intensity scale. Confirm whether smoother device interiors come at the expense of the thin dark line.

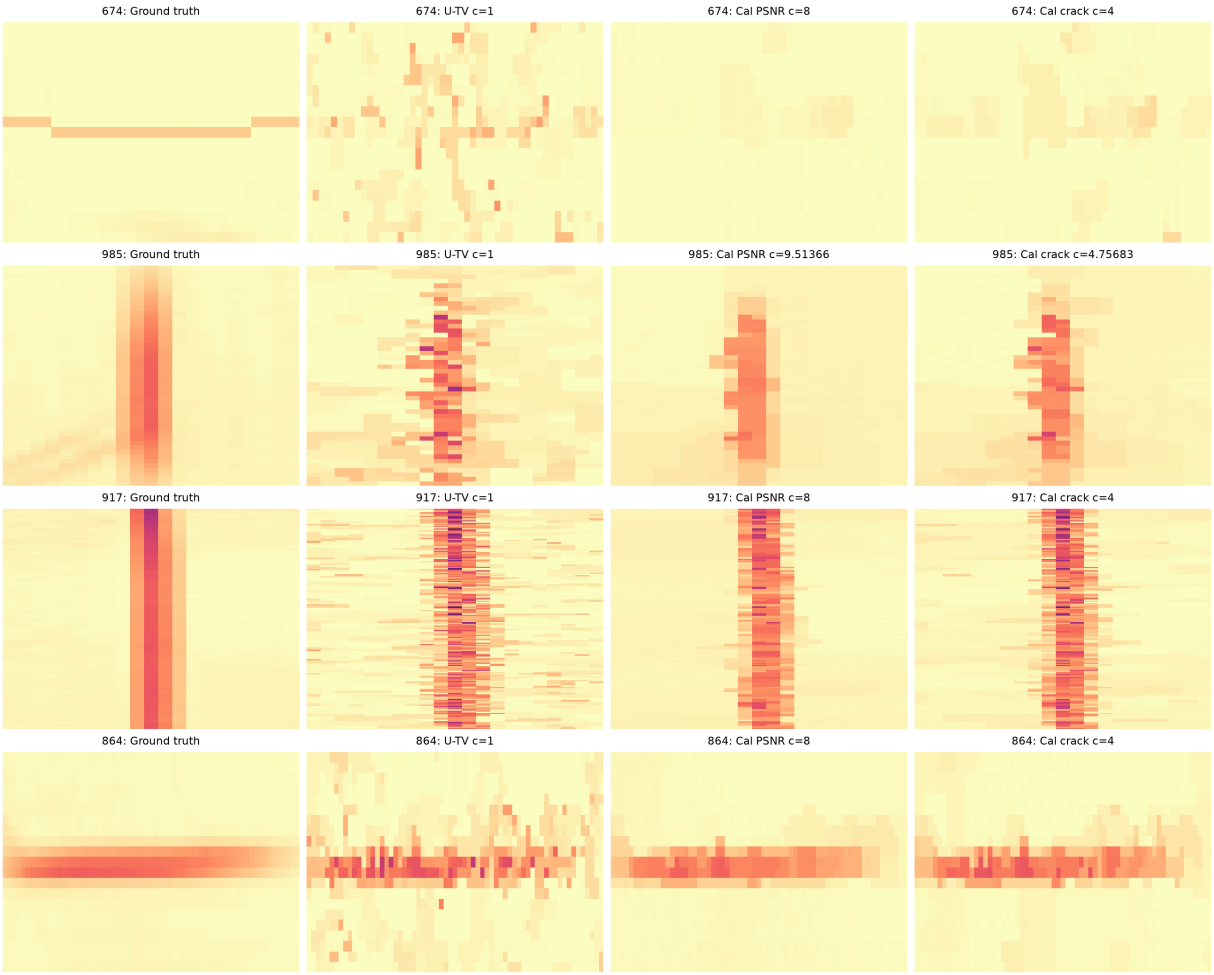


Figure 4: Initial fine-line ROI overview. Show identical line-centred crop coordinates with a shared colour scale. The fixed-location line metric is a diagnostic, not a validated automatic crack detector.

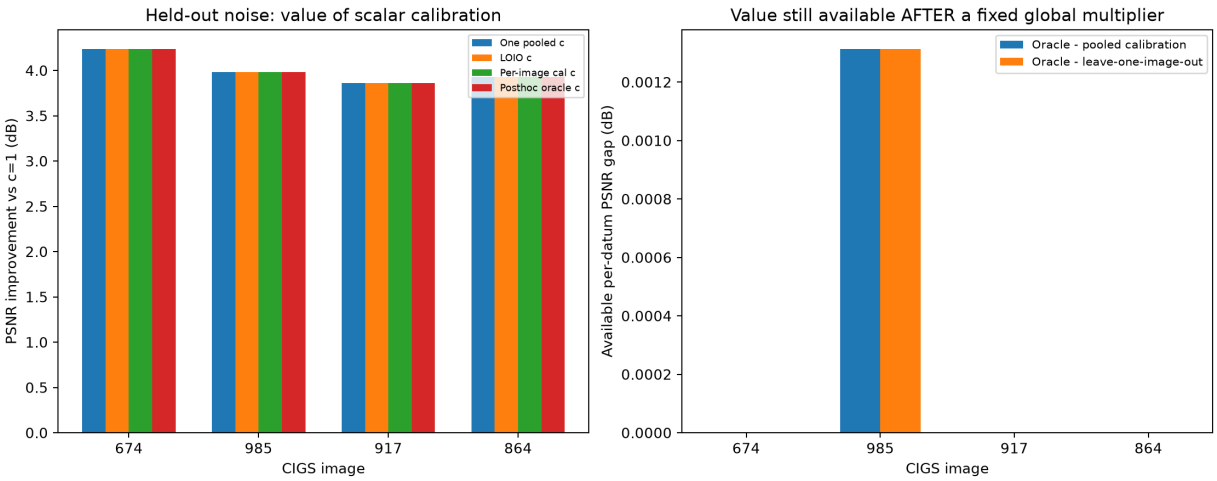


Figure 5: Does a per-datum oracle improve beyond a fixed multiplier?. Left: gains over c=1 on held-out noise. Right: upper-bound oracle minus pooled/LOIO calibrated scale, the most useful measure for deciding whether a learned scaling head is worthwhile.

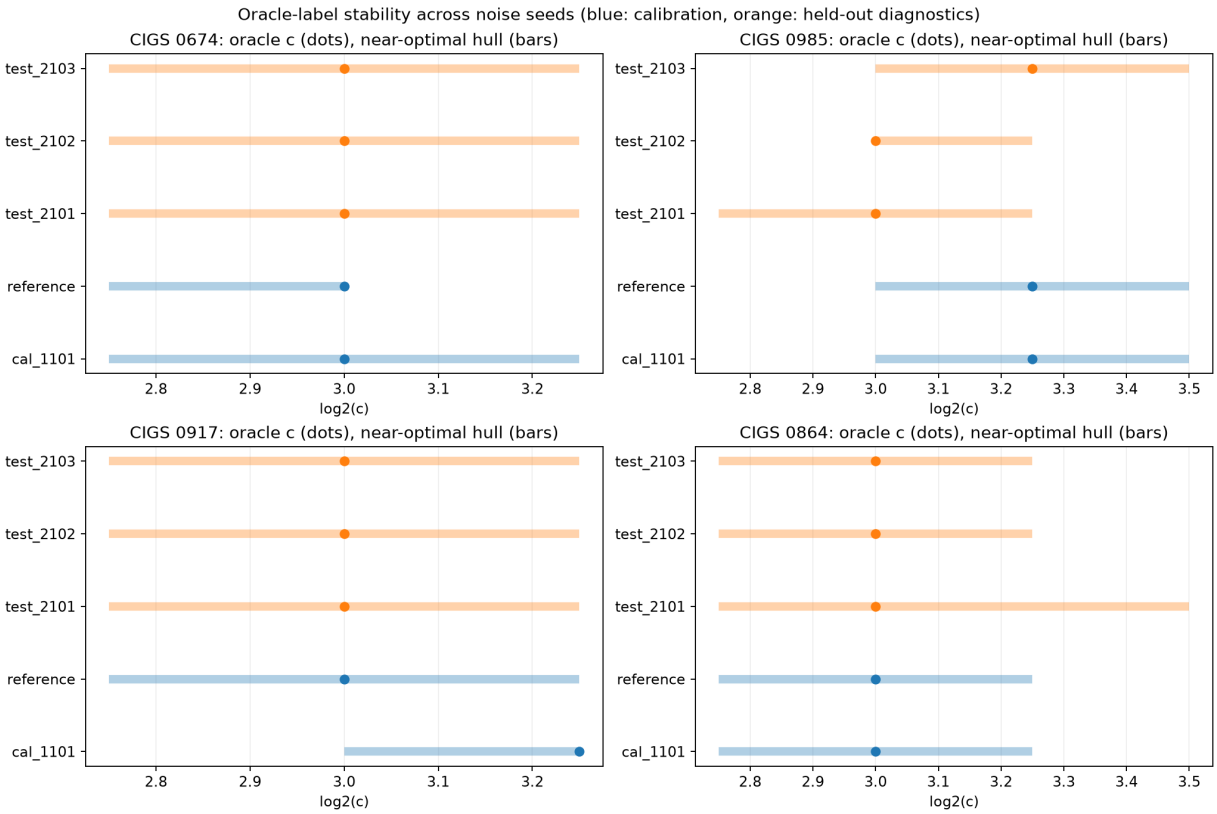


Figure 6: Oracle log-scales and near-optimal sets. Each dot is an optimal scale for one noise realisation; the bar is only the min/max hull of the sampled near-optimal set. Large changes in labels with tiny regret indicate unstable regression targets.

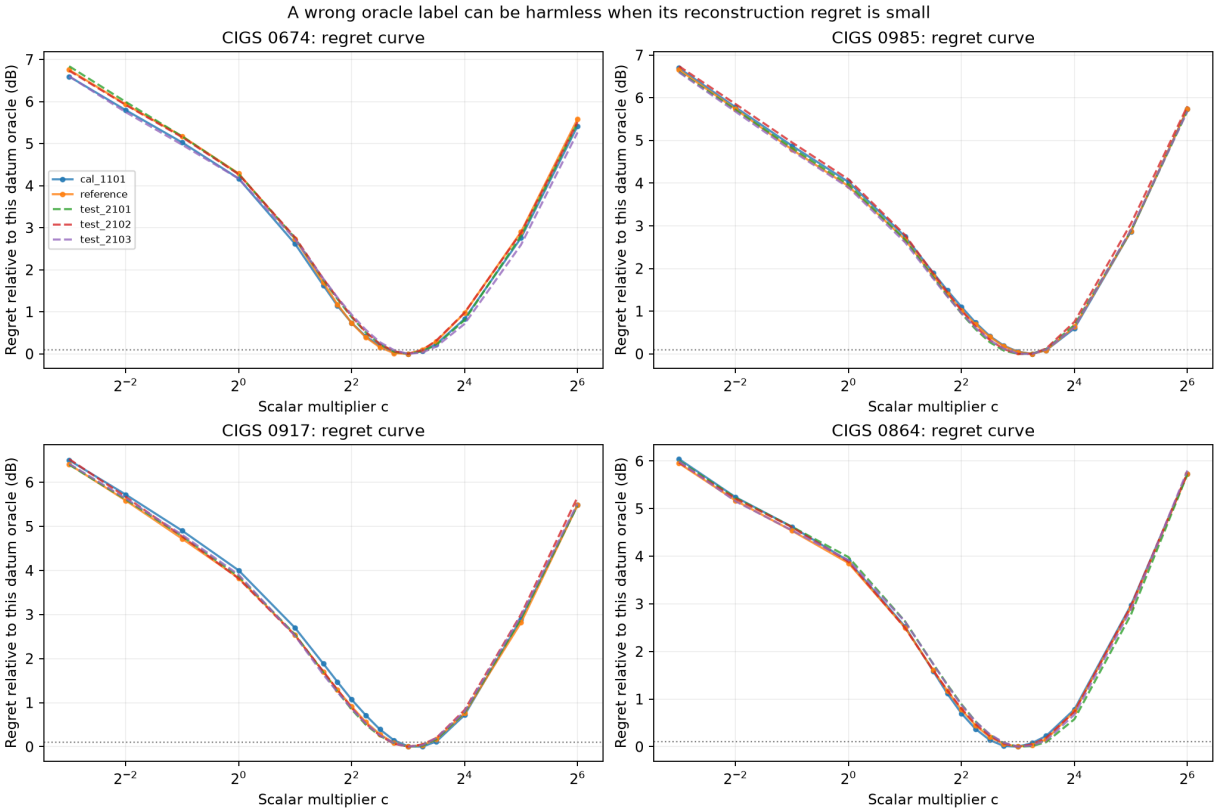


Figure 7: Per-datum PSNR regret across scales. Compare the performance loss of choosing the wrong scale. Hard scale-label errors are less important when the PSNR curve has a broad flat maximum.

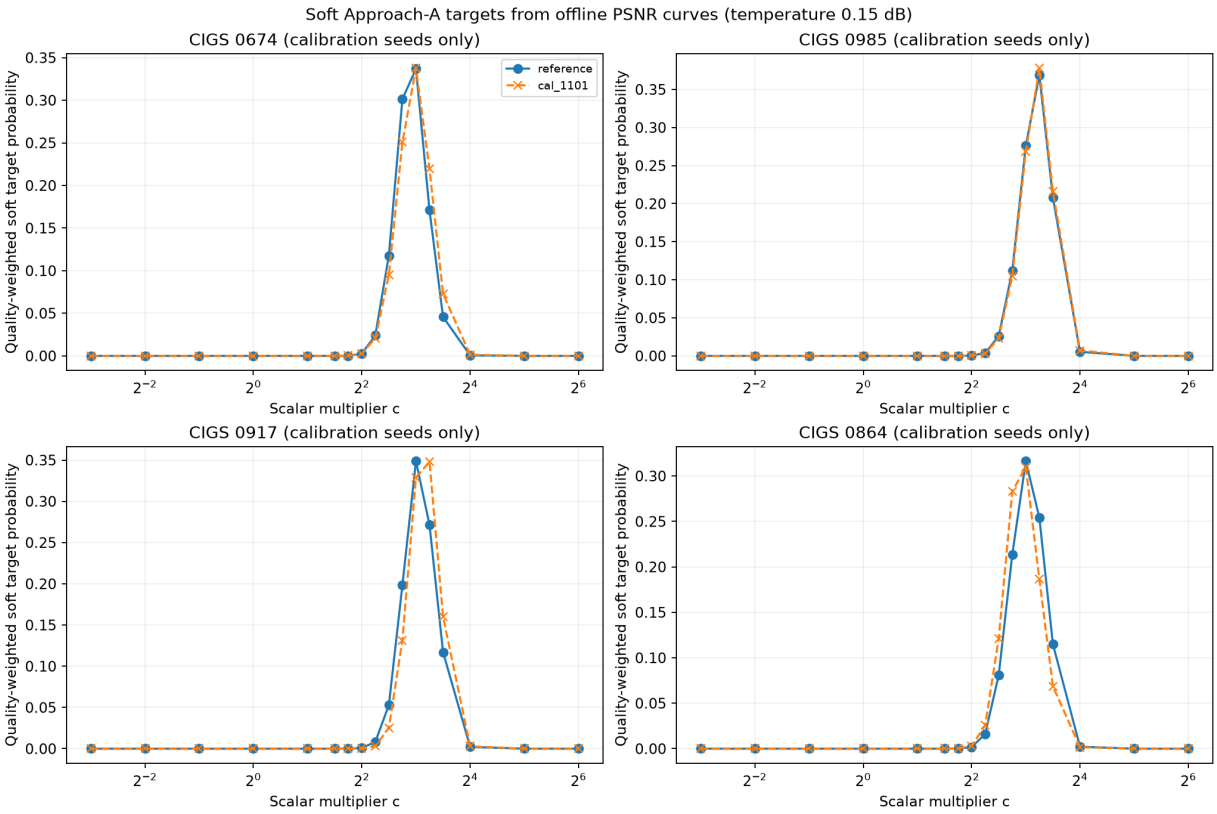


Figure 8: Soft oracle training targets on calibration data. Targets put substantial probability on all nearly optimal tested scales. All plotted probabilities are computed offline using calibration ground truth, not from a learned predictor.

6 Detailed held-out visual atlas: all images and test noise seeds

Each of the four CIGS images and each of the three held-out test noise realisations receives a full-image and enlarged annotated-line ROI panel. Every panel compares ground truth, $c = 1$, pooled scale, leave-one-image-out scale, per-image PSNR scale, line-aware calibration scale, and the test-ground-truth posthoc oracle. All panels use a common display range of [0,1] without modifying the saved float32 arrays.

6.1 CIGS 0674

The verified line is horizontal with nominal width 1 pixel(s); centre coordinate 115, longitudinal indices [106,139). Crack annotations were audited against the original PNG before executing the sweep.

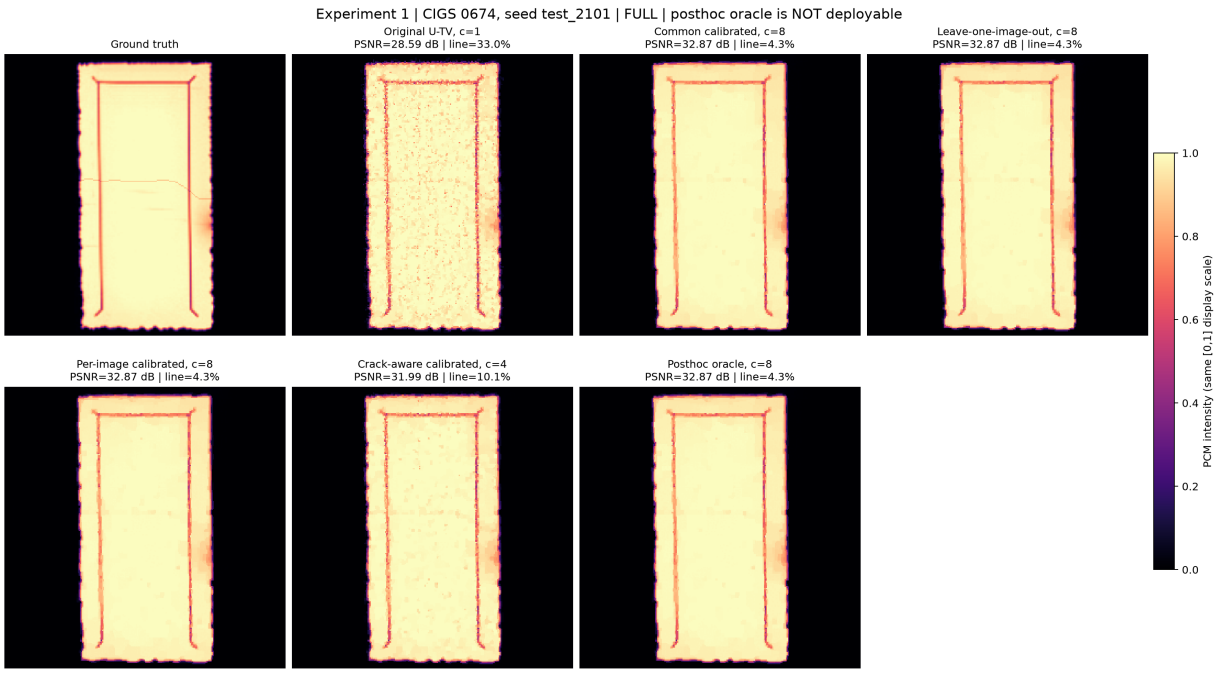


Figure 9: CIGS 0674, test_2101: full reconstructions. On held-out noise seed test_2101, the posthoc oracle c=8.0000 has PSNR 32.867 dB; its gain over c=1 is 4.279 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

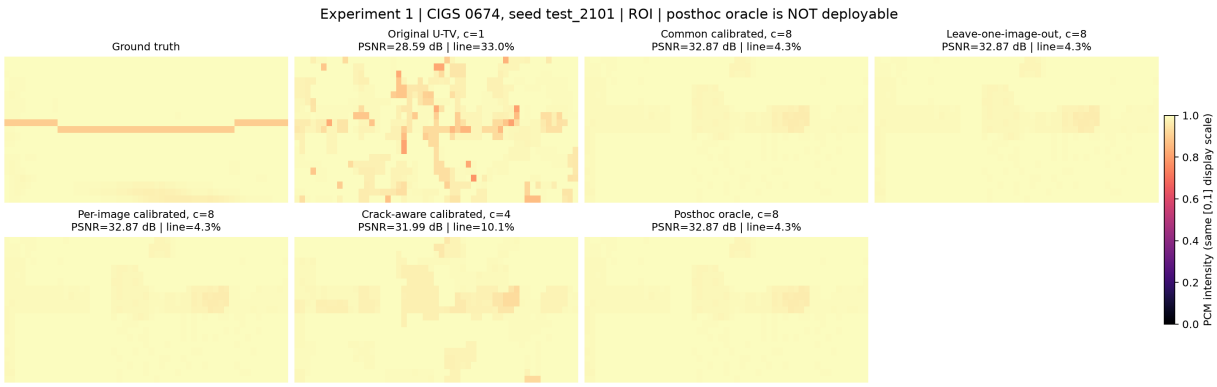


Figure 10: CIGS 0674, test_2101: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2101, the posthoc oracle c=8.0000 has PSNR 32.867 dB; its gain over c=1 is 4.279 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

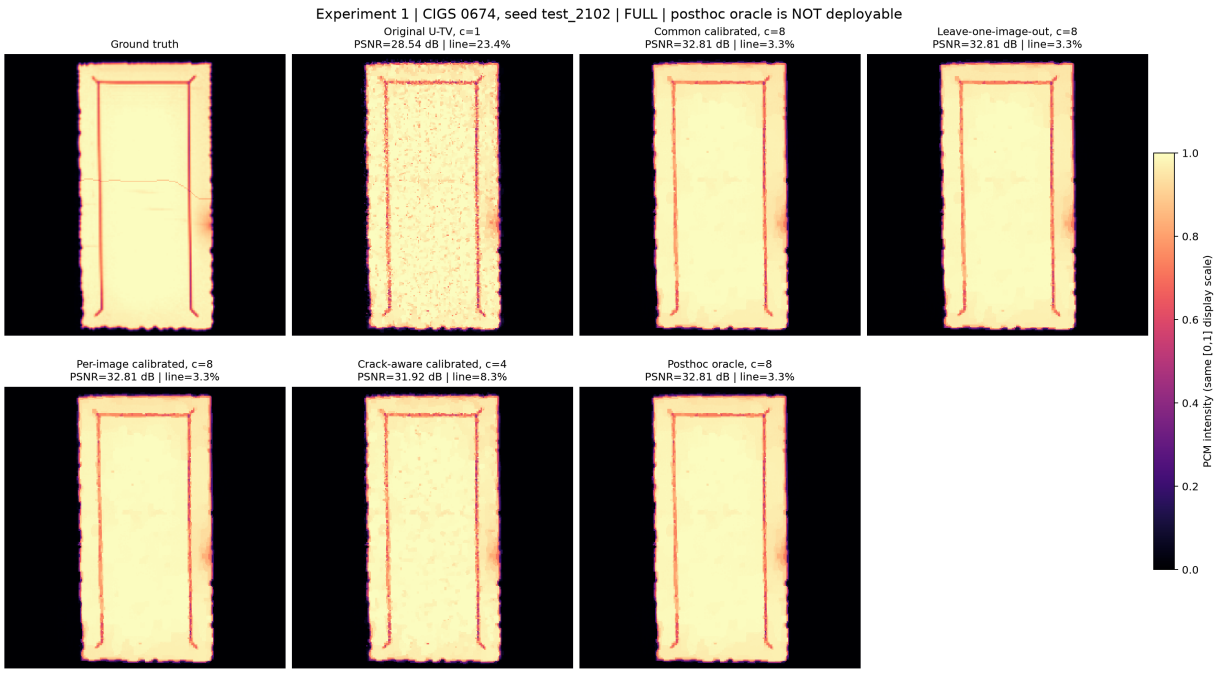


Figure 11: CIGS 0674, test_2102: full reconstructions. On held-out noise seed test_2102, the posthoc oracle c=8.0000 has PSNR 32.813 dB; its gain over c=1 is 4.271 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

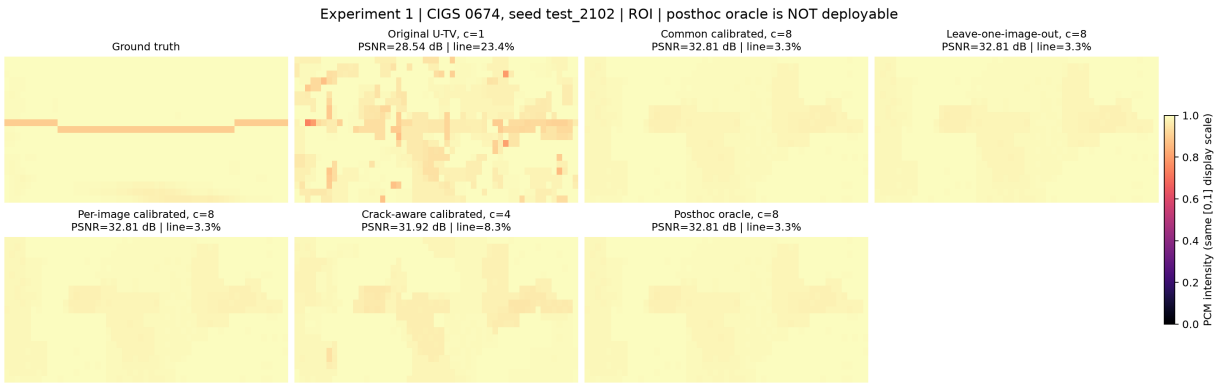


Figure 12: CIGS 0674, test_2102: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2102, the posthoc oracle c=8.0000 has PSNR 32.813 dB; its gain over c=1 is 4.271 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

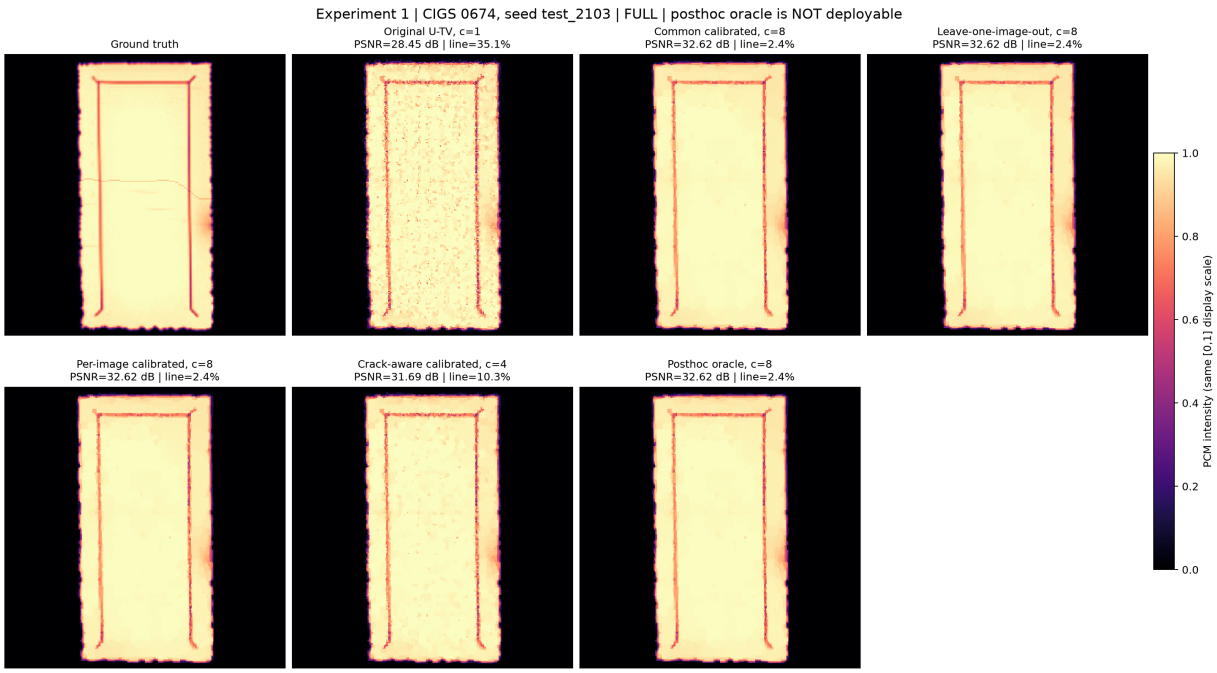


Figure 13: CIGS 0674, test_2103: full reconstructions. On held-out noise seed test_2103, the posthoc oracle c=8.0000 has PSNR 32.622 dB; its gain over c=1 is 4.168 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

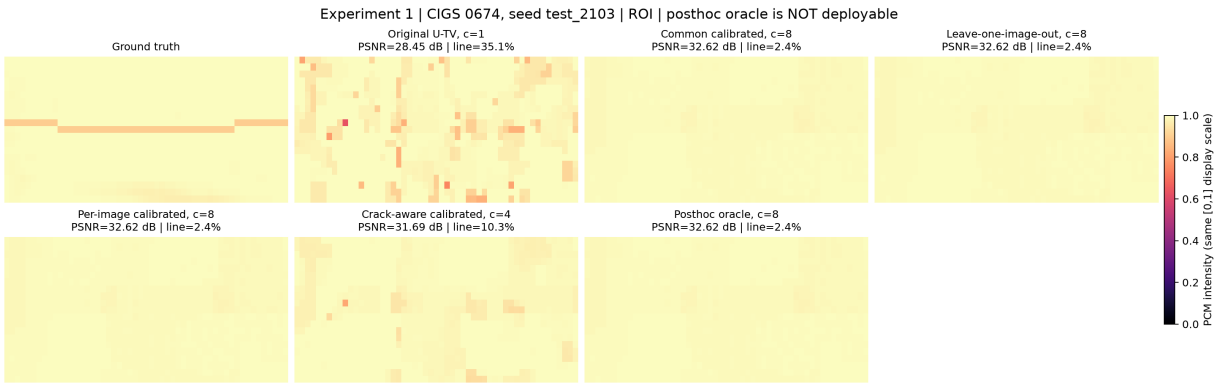


Figure 14: CIGS 0674, test_2103: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2103, the posthoc oracle c=8.0000 has PSNR 32.622 dB; its gain over c=1 is 4.168 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

6.2 CIGS 0985

The verified line is vertical with nominal width 3 pixel(s); centre coordinate 107, longitudinal indices [135,168). Crack annotations were audited against the original PNG before executing the sweep.

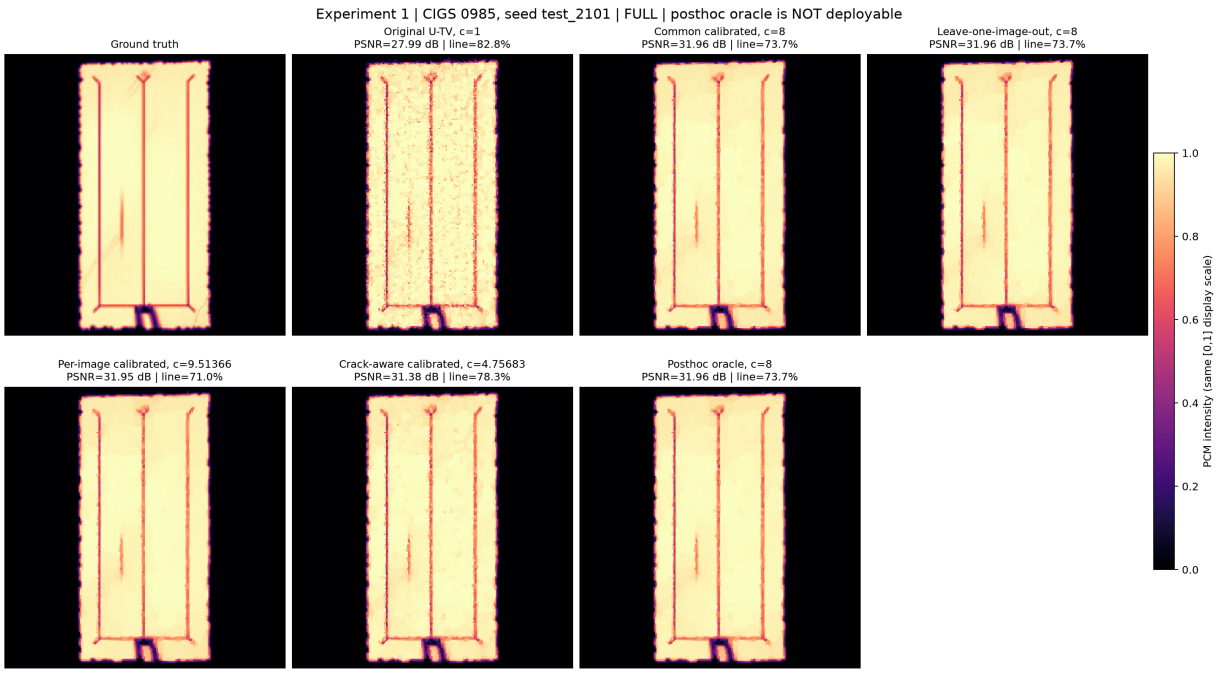


Figure 15: CIGS 0985, test_2101: full reconstructions. On held-out noise seed test_2101, the posthoc oracle c=8.0000 has PSNR 31.956 dB; its gain over c=1 is 3.966 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

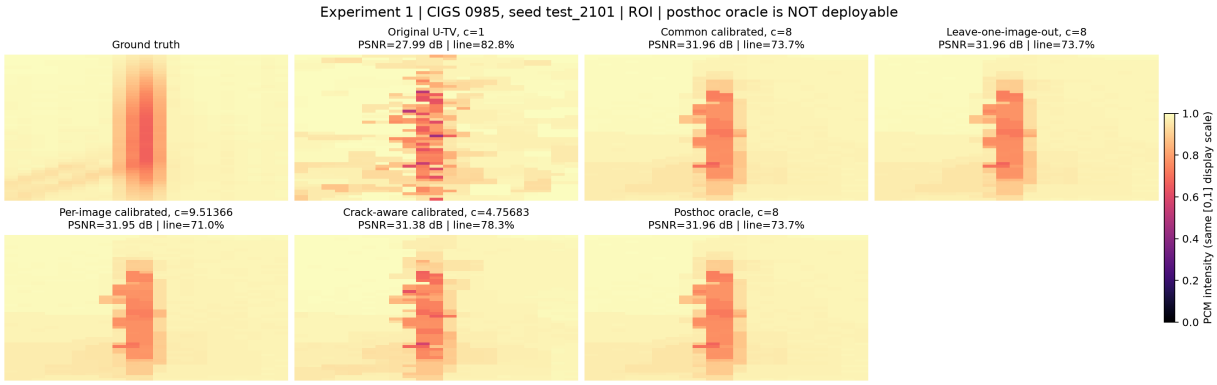


Figure 16: CIGS 0985, test_2101: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2101, the posthoc oracle c=8.0000 has PSNR 31.956 dB; its gain over c=1 is 3.966 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

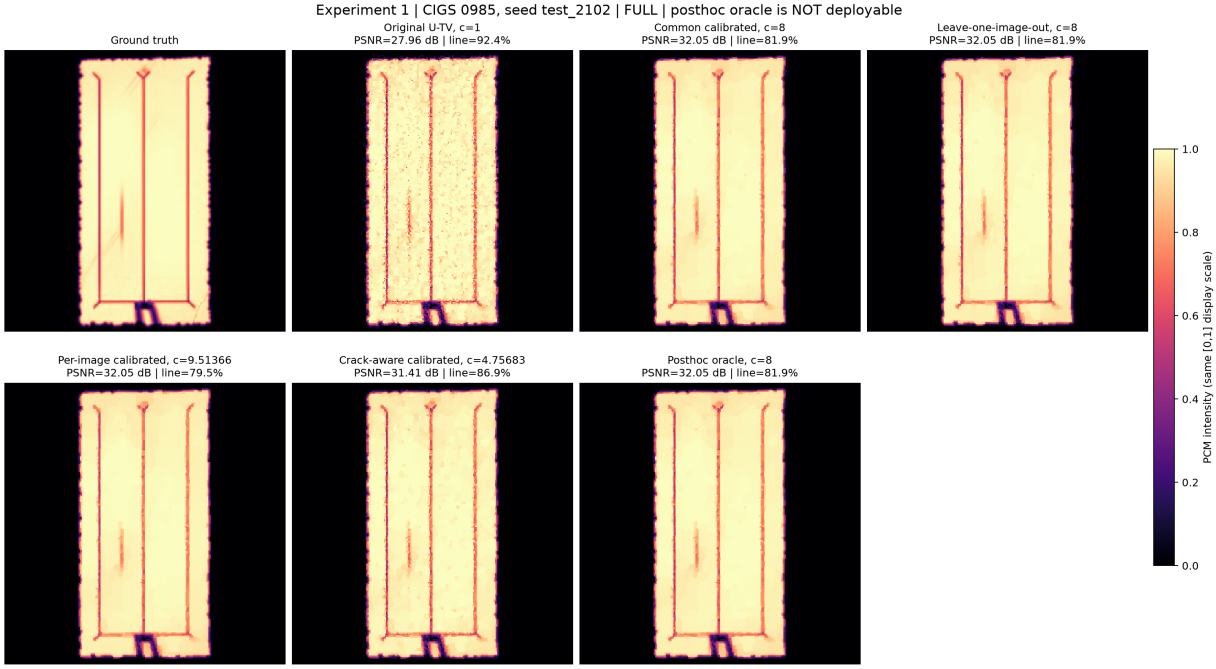


Figure 17: CIGS 0985, test_2102: full reconstructions. On held-out noise seed test_2102, the posthoc oracle c=8.0000 has PSNR 32.048 dB; its gain over c=1 is 4.089 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

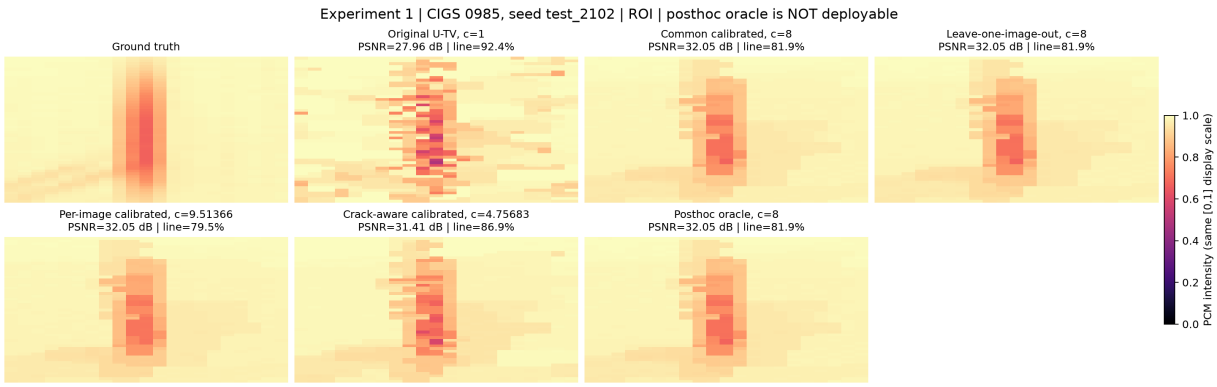


Figure 18: CIGS 0985, test_2102: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2102, the posthoc oracle c=8.0000 has PSNR 32.048 dB; its gain over c=1 is 4.089 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

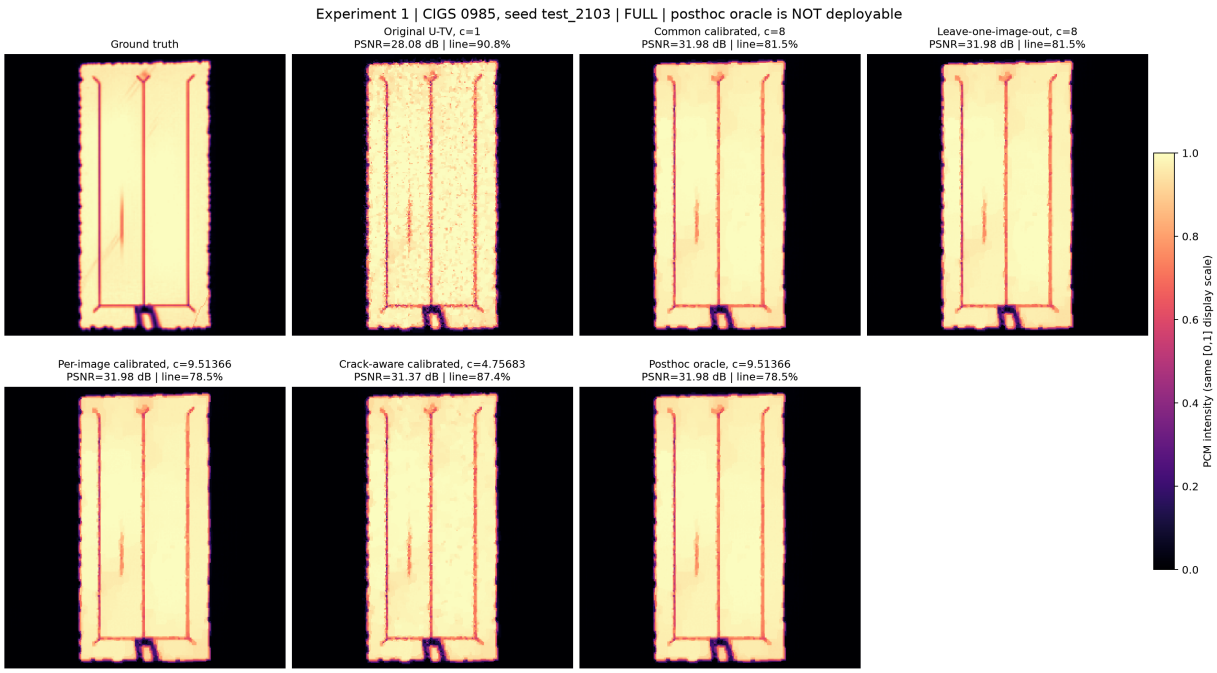


Figure 19: CIGS 0985, test_2103: full reconstructions. On held-out noise seed test_2103, the posthoc oracle c=9.5137 has PSNR 31.980 dB; its gain over c=1 is 3.899 dB, and its extra gain over a single pooled c is 0.004 dB. Near-optimal discrete scales (within 0.10 dB) are 8.0;9.51365692;11.313708499. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

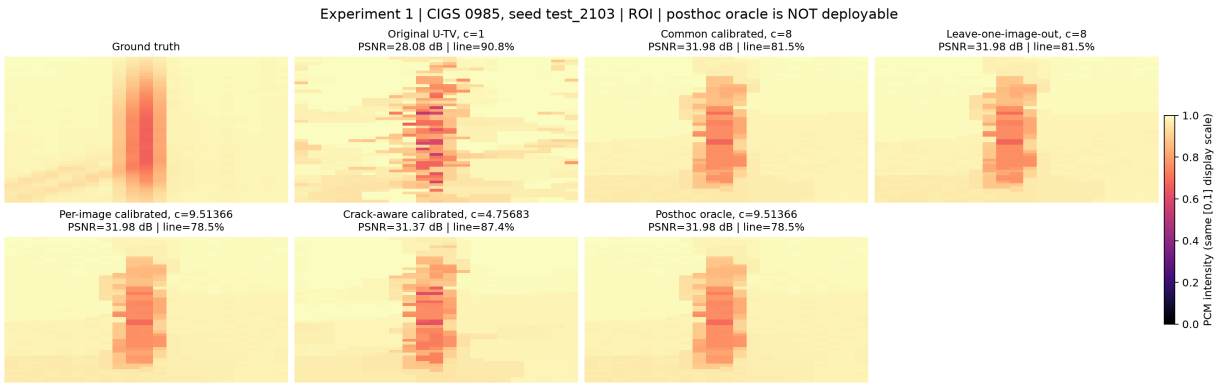


Figure 20: CIGS 0985, test_2103: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2103, the posthoc oracle c=9.5137 has PSNR 31.980 dB; its gain over c=1 is 3.899 dB, and its extra gain over a single pooled c is 0.004 dB. Near-optimal discrete scales (within 0.10 dB) are 8.0;9.51365692;11.313708499. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

6.3 CIGS 0917

The verified line is vertical with nominal width 3 pixel(s); centre coordinate 88, longitudinal indices [40,213). Crack annotations were audited against the original PNG before executing the sweep.

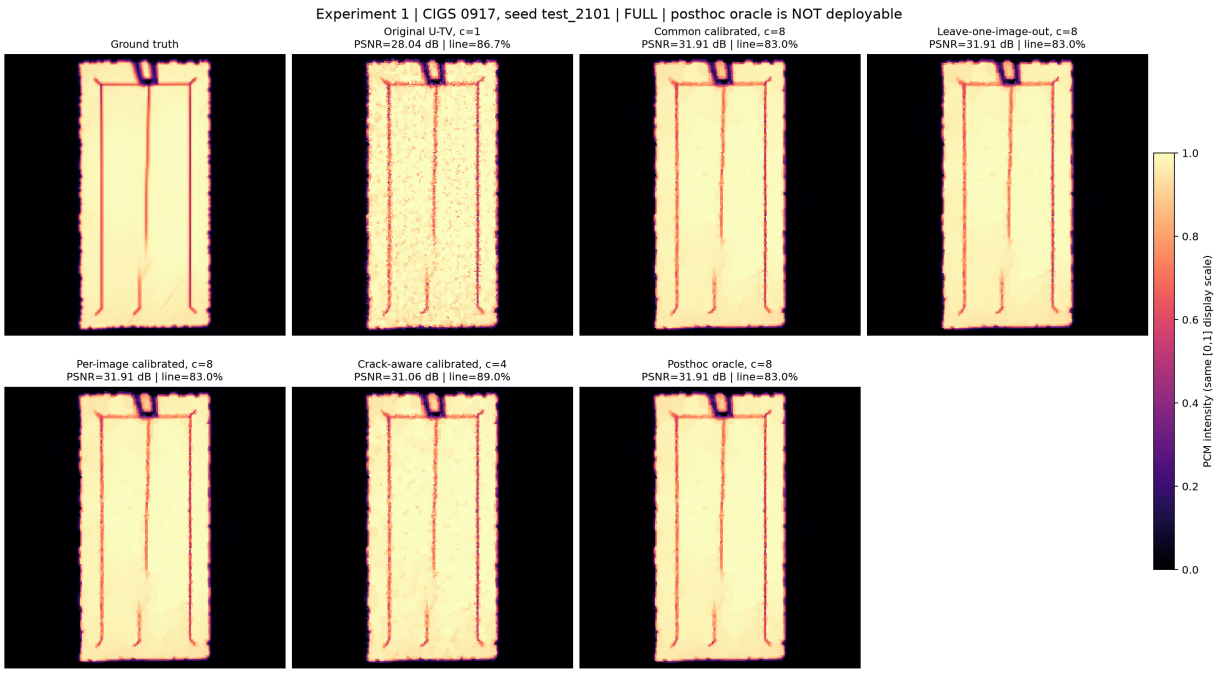


Figure 21: CIGS 0917, test_2101: full reconstructions. On held-out noise seed test_2101, the posthoc oracle c=8.0000 has PSNR 31.911 dB; its gain over c=1 is 3.867 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

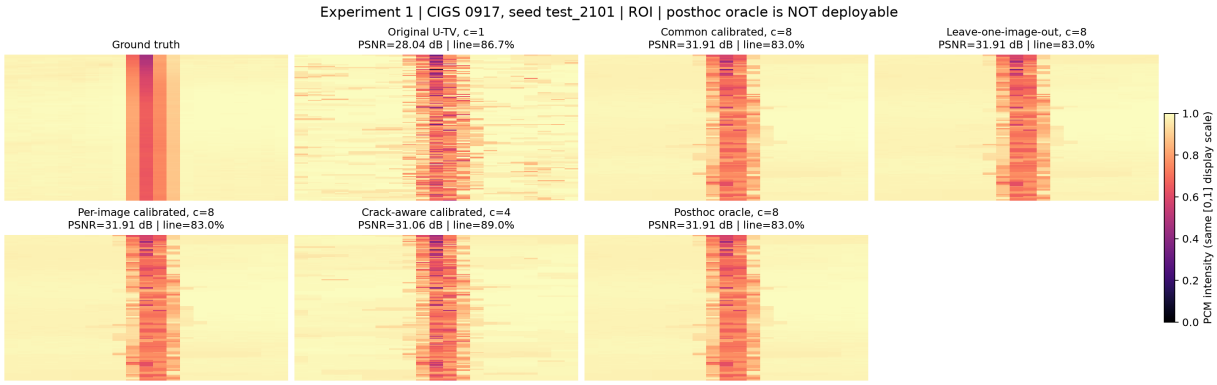


Figure 22: CIGS 0917, test_2101: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2101, the posthoc oracle c=8.0000 has PSNR 31.911 dB; its gain over c=1 is 3.867 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

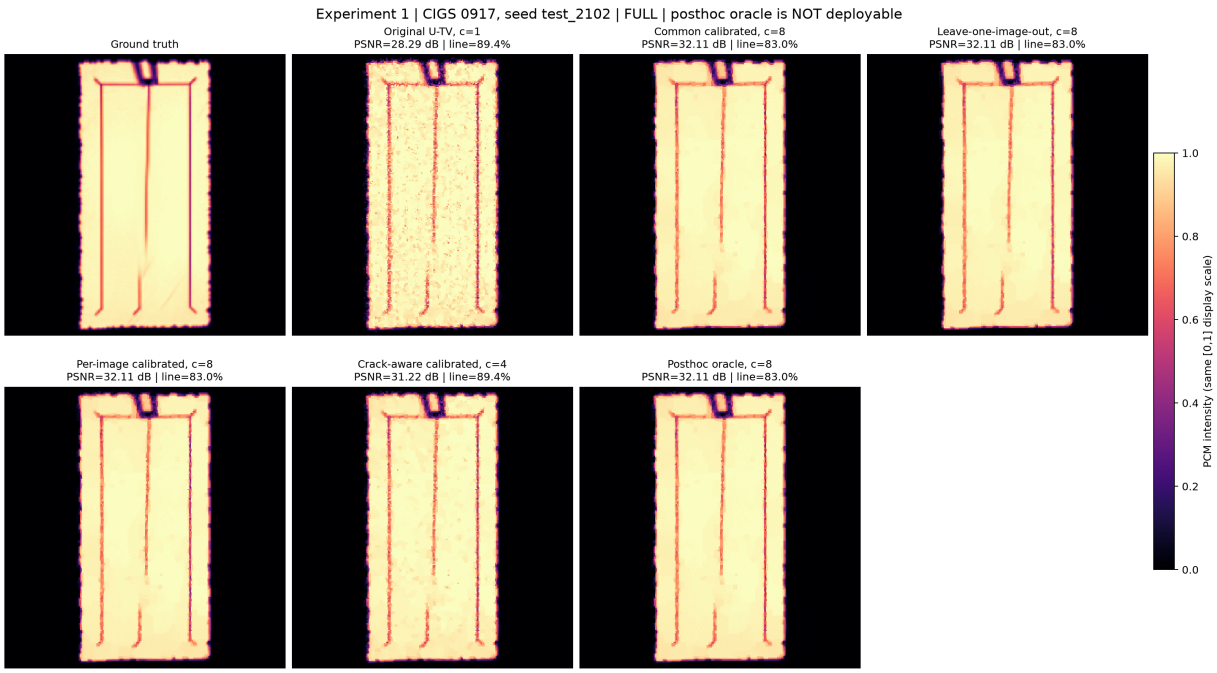


Figure 23: CIGS 0917, test_2102: full reconstructions. On held-out noise seed test_2102, the posthoc oracle c=8.0000 has PSNR 32.106 dB; its gain over c=1 is 3.816 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

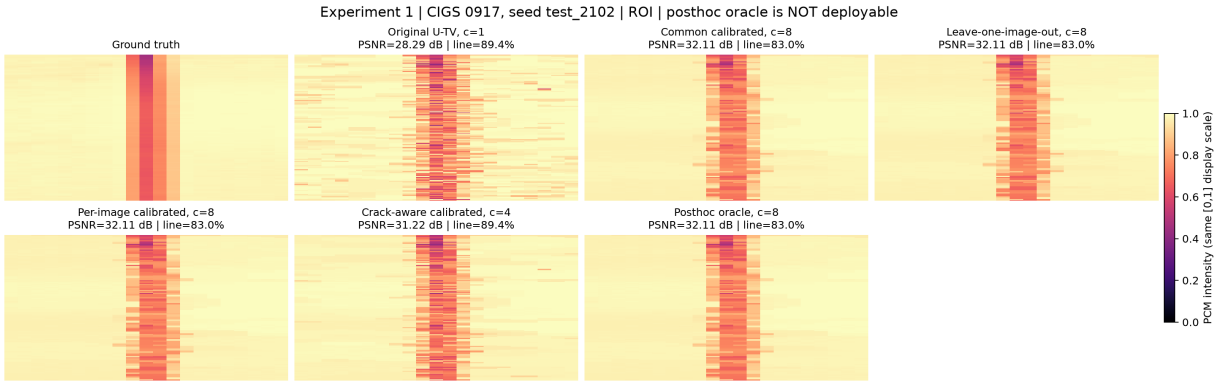


Figure 24: CIGS 0917, test_2102: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2102, the posthoc oracle c=8.0000 has PSNR 32.106 dB; its gain over c=1 is 3.816 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

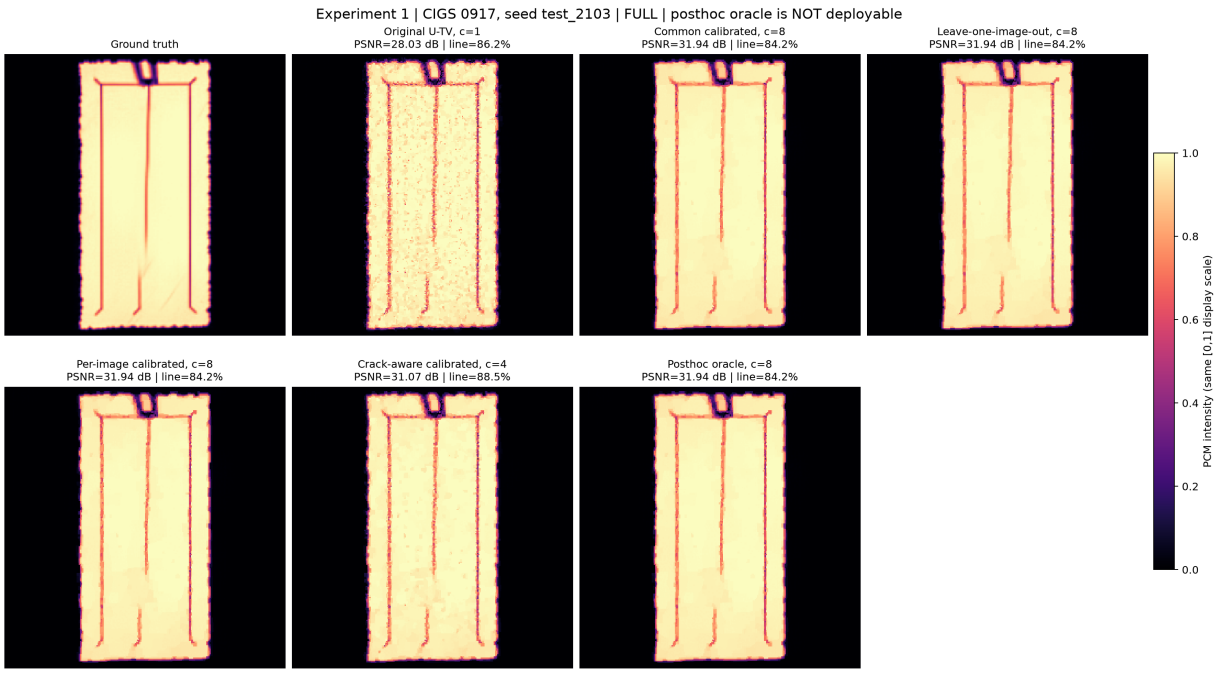


Figure 25: CIGS 0917, test_2103: full reconstructions. On held-out noise seed test_2103, the posthoc oracle c=8.0000 has PSNR 31.938 dB; its gain over c=1 is 3.903 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

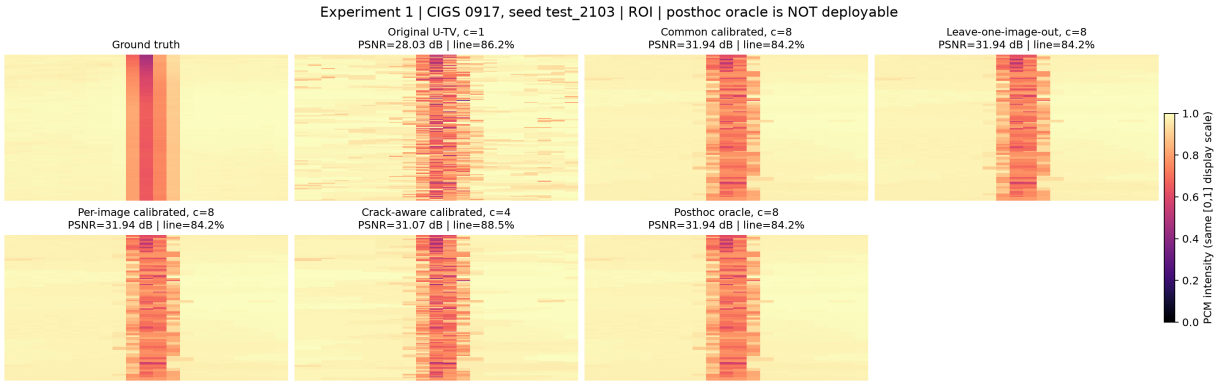


Figure 26: CIGS 0917, test_2103: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2103, the posthoc oracle c=8.0000 has PSNR 31.938 dB; its gain over c=1 is 3.903 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

6.4 CIGS 0864

The verified line is horizontal with nominal width 3 pixel(s); centre coordinate 132, longitudinal indices [77,126). Crack annotations were audited against the original PNG before executing the sweep.

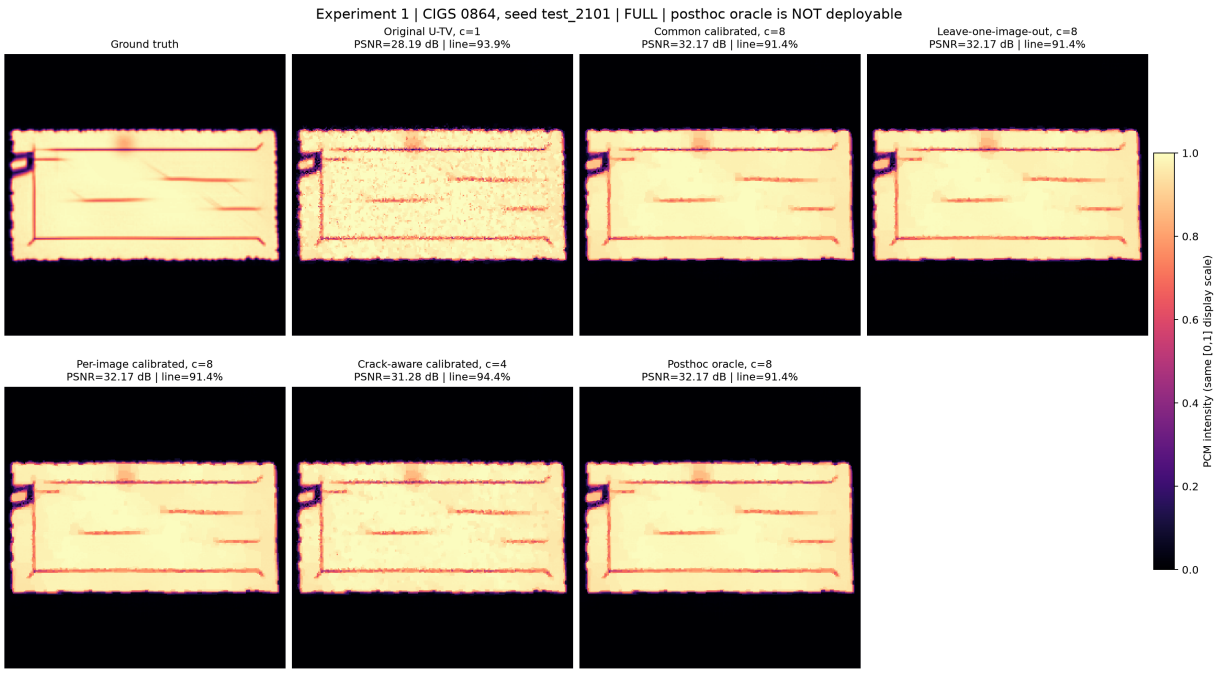


Figure 27: CIGS 0864, test_2101: full reconstructions. On held-out noise seed test_2101, the posthoc oracle c=8.0000 has PSNR 32.171 dB; its gain over c=1 is 3.982 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692;11.313708499. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

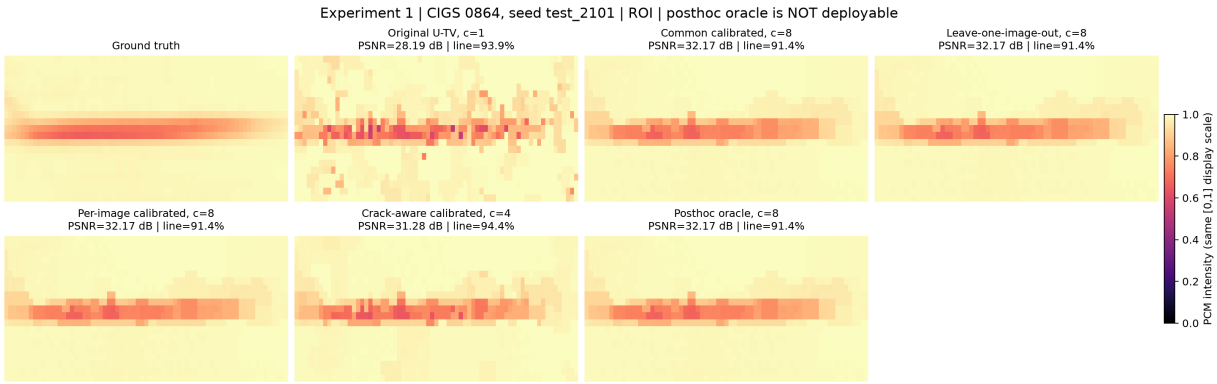


Figure 28: CIGS 0864, test_2101: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2101, the posthoc oracle c=8.0000 has PSNR 32.171 dB; its gain over c=1 is 3.982 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692;11.313708499. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

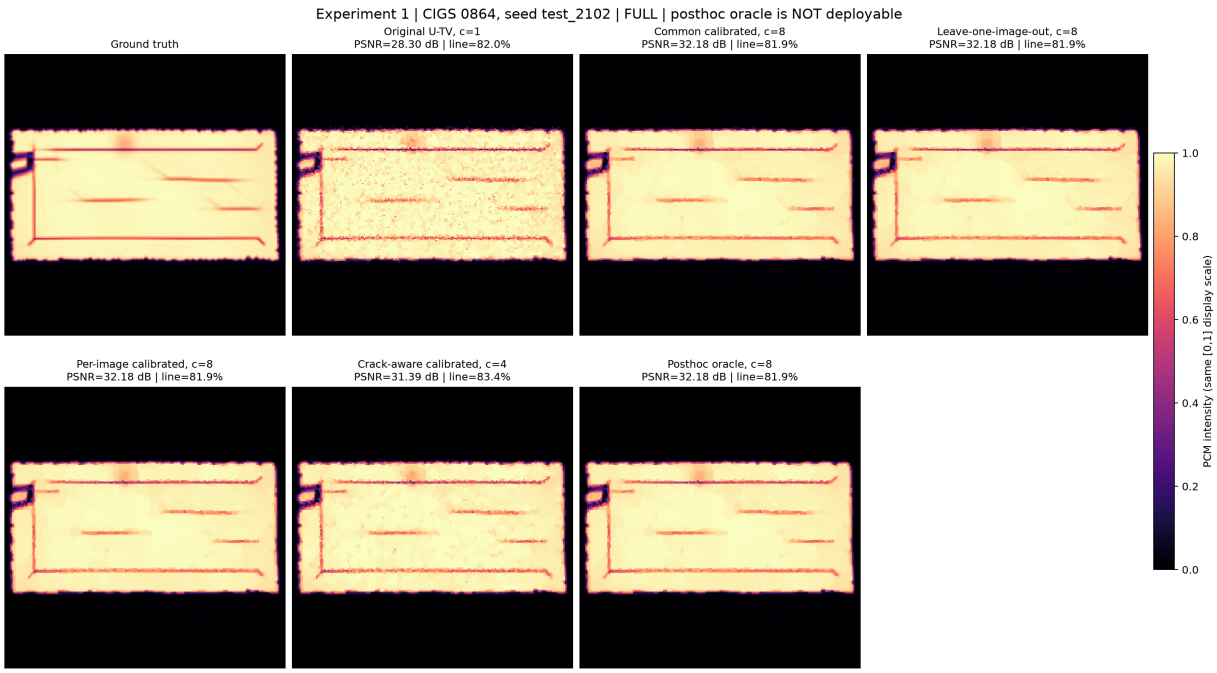


Figure 29: CIGS 0864, test_2102: full reconstructions. On held-out noise seed test_2102, the posthoc oracle c=8.0000 has PSNR 32.183 dB; its gain over c=1 is 3.887 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

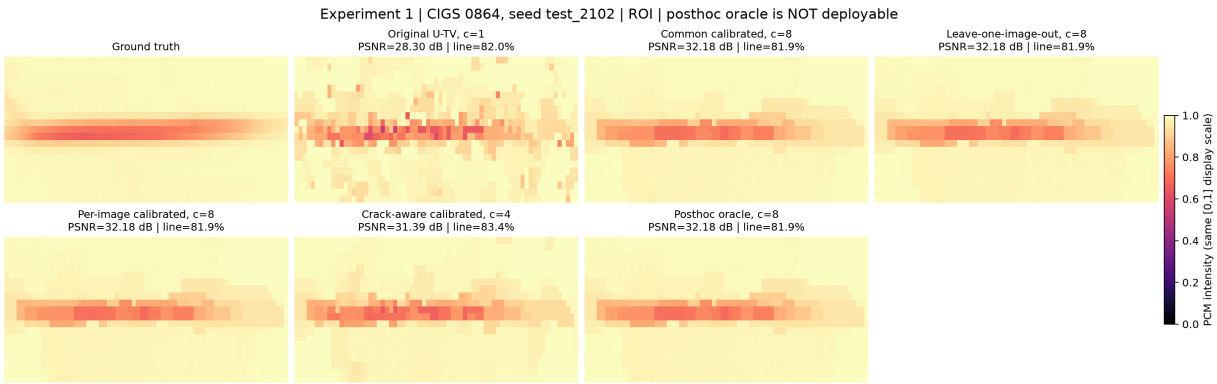


Figure 30: CIGS 0864, test_2102: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2102, the posthoc oracle c=8.0000 has PSNR 32.183 dB; its gain over c=1 is 3.887 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

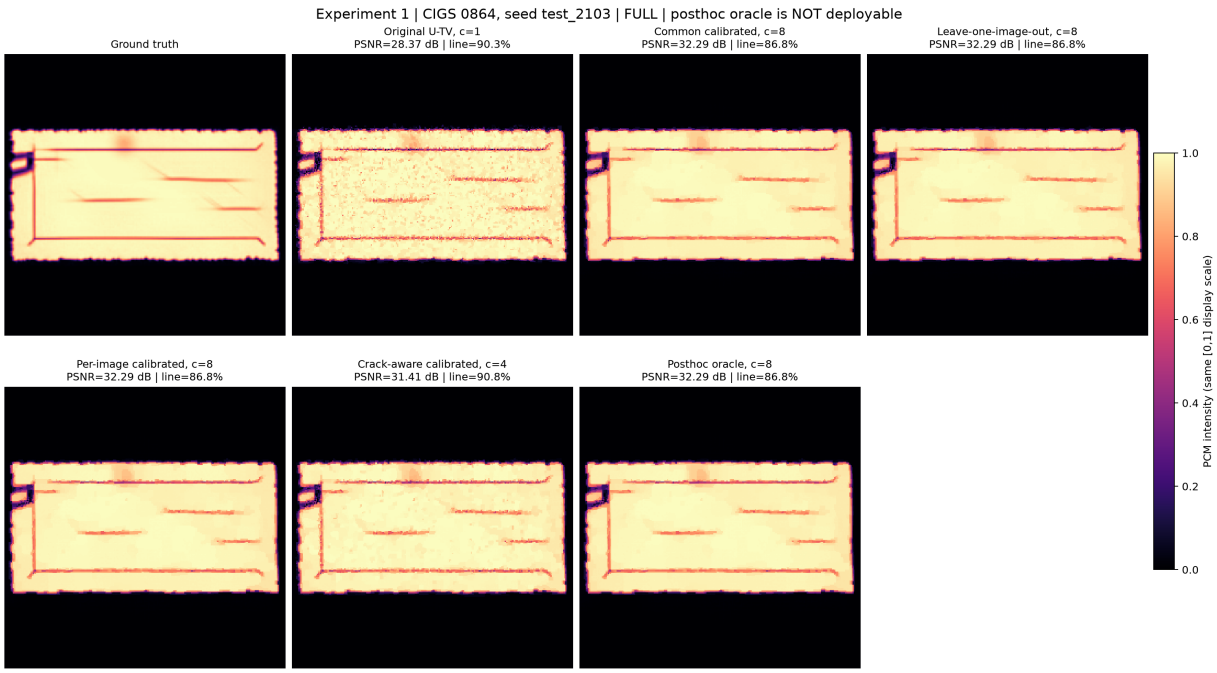


Figure 31: CIGS 0864, test_2103: full reconstructions. On held-out noise seed test_2103, the posthoc oracle c=8.0000 has PSNR 32.290 dB; its gain over c=1 is 3.919 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone.

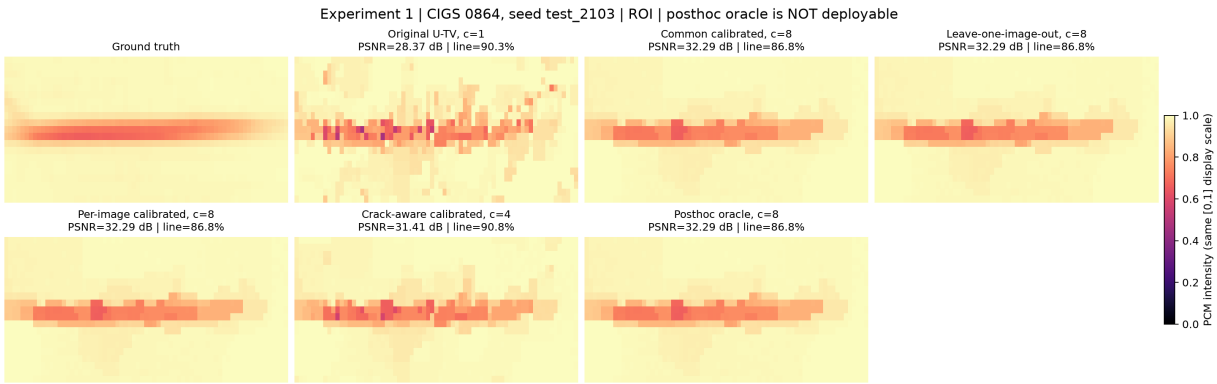


Figure 32: CIGS 0864, test_2103: enlarged thin-line ROI. This is the same measurement and the same ordered methods as the preceding full image. On held-out noise seed test_2103, the posthoc oracle c=8.0000 has PSNR 32.290 dB; its gain over c=1 is 3.919 dB, and its extra gain over a single pooled c is 0.000 dB. Near-optimal discrete scales (within 0.10 dB) are 6.727171322;8.0;9.51365692. Whether the line survives should be judged from the ROI and signed line-contrast scores, not PSNR alone. Some annotated lines are electrode tracks rather than confirmed material defects; false structures must also be checked.

7 Oracle-label stability and limitations

CIGS	Seeds	SD(log2 c*)	log2 c* span	Near-opt set overlap
0674	5	0.000	0.000	0.667
0985	5	0.137	0.250	0.500
0917	5	0.112	0.250	0.667
0864	5	0.000	0.000	0.750

The oracle label is a function of both the noisy datum AND the unknown ground truth. Two candidate scenes can produce similar undersampled measurements yet have different best scales. Therefore an arbitrary improvement by the posthoc oracle cannot automatically be learned from measurements. This Stage I experiment contains just four independent image structures and cannot measure neural generalisation. The prepared Stage II trains the scale head on exactly 50 other image structures (10 per category), and selects it using 25 separate validation images. The same four CIGS images are later used with NEW noise seeds for a small, non-blinded proof of concept.

8 Conclusions and staged next experiments

The most important quantity for deciding on Approach A is the *additional* oracle improvement beyond one validation-calibrated global multiplier. If this is small, a fixed multiplier is preferable. If the remaining gap is large but oracle labels change abruptly across noise seeds while quality curves are flat, train on near-optimal regions or soft metric-curve targets rather than a hard log-scale label. Only after an image-disjoint dataset is available should we freeze the map network and train the scalar head.

1. Verify the full original images and manually approve all line locations, with pixel hashes.
2. Repeat with an independently chosen checkpoint trained on simulated PCM noise; keep the old checkpoint as historical control.
3. Compare c=1, pooled calibration c, LOIO c, per-image c, and posthoc per-datum oracle on held-out seeds.
4. Quantify label instability, near-optimal sets, soft targets and quality regret; keep crack preservation as a separate objective.
5. Use the 50 independent training images and 25 validation images from the five synthetic PCM categories to train/select Approach A scalar predictors without using these four images.
6. Run final evaluation with three NEW noise seeds on the same four CIGS structures, excluding them entirely from head training and validation; acknowledge Stage I has already exposed these structures.

9 Artifact provenance and numerical integrity

All numerical scores are read directly from full-precision CSV metrics and stored float32 arrays. Figures are display-only resamplings; they are not used to calculate metrics. The original measured arrays, sampled indices, noise realisations, learned maps and checkpoints remain in the immutable experiment output directory. TeX and PDF are generated after complete coarse/refinement sweeps, never before.

Protocol identifier: experiment_1_on_28-10-2026_learned_posttraining_scale_calibration_v3
Code revision: ce5dcb4780780d9779eee6c43117d73d001db4af
Run directory: /store/LION/ttv22/pcm/results/experiment_1_on_28-10-2026/old
Checkpoint: /store/LION/ttv22/pcm/results/pcm/pcm_supervised/20260920_025657_TV/checkpoints/best_